ELE529E Embedded Systems Project Progress Report

Course: ELE529E Embedded Systems
Submission Date: [30/04/2026]

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1. Project Overview

Project Title	**[Intelligent Dual-Axis Solar Tracker with Predictive Filtering]**
Team Members	[Makbule Ozge Ozler] ([704252003]), [Mehmet Furkan Kalem(unofficially registered)] ([504251293])
Project Start Date	[13/03/2026]
Expected Completion	[22/05/2026]

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2. Project Milestones & Delivery Plan

Milestone	**Tasks**	**Deadline**	**Status (✓/✕)**
1. Requirement Analysis	Define project scope, objectives, constraints and STM32 peripheral selection.	[20/03/2026]	✕
2. System Design	Hardware/software architecture, UML diagrams FreeRTOS task prioritization, Pinout.	[10/04/2026]	✕
3. Prototype Development	Implement core functionalities (Sensor drivers (ADC, I2C), PWM control for servos.).	[30/04/2026]	✕
4. Testing & Debugging	Unit tests, system validation, performance analysis. Filtering optimization, power consumption profiling.	[15/05/2025]	✕
5. Final Demo & Report	Final system integration and demo video preparation. Prepare presentation, submit final report.	[22/05/2026]	✕

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3. Individual Contribution Plans

Student 1: [Makbule Ozge Ozler]

Task	Responsibility	Estimated Time
Software Architecture	Designing the system state-machine and UML logic flow.	2 weeks
Data Processing	Implementing predictive filtering (Kalman or EMA) for LDR data.	2 weeks
Control Algorithms	Logic analyzer, power profiling.	1 week
Cloud Integration	Managing UART communication for ESP8266 & ThingSpeak.	1 week

Student 2: [Mehmet Furkan Kalem(unofficially registered)]

Task	Responsibility	Estimated Time
Hardware Setup	STM32 configuration, sensor interfacing. Peripheral initialization (ADC, I2C, PWM) on STM32F410.	2 weeks
Low-Level Drivers	Custom INA219 driver development and servo PWM logic.	2 weeks
Real-Time Scheduling	FreeRTOS task creation, Resource Protection (Mutex/Semaphores).	1.5 week
Optimization	Power-aware coding and hardware-level performance tuning.	1 week

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4. Project Quality Utility Tree

Objective: Ensure reliability, performance, and usability.

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1. Functional Correctness
   ├── 1.1 Positioning Accuracy (±1° error margin for dual-axis)
   ├── 1.2 Power Monitoring (INA219 precision <1%)
   └── 1.3 Power Efficiency (Energy generated > System consumption)

2. Robustness
   ├── 2.1 Environmental Noise Handling (Predictive filtering on LDRs)
   └── 2.2 Fault Recovery (Re-calibration after power loss)

3. Maintainability
   ├── 3.1 Modular Code (Separation of Hardware Abstraction Layer - HAL)
   └── 3.2 Real-Time Monitoring (Live telemetry via ThingSpeak)
```

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5. Project Structure Diagram (PlantUML)

```
@startuml
package "Hardware Layer" {
  [STM32F410RB] --> [LDR Sensors (ADC)]
  [STM32F410RB] --> [INA219 (I2C)]
  [STM32F410RB] --> [2x Servo Motors (PWM)]
  [STM32F410RB] --> [ESP8266 (UART)]
}

package "Software Layer (FreeRTOS)" {
  [FreeRTOS] --> [Task 1: Data Acquisition]
  [FreeRTOS] --> [Task 2: Control Logic]
  [FreeRTOS] --> [Task 3: Communication]
}

[ThingSpeak] <.. [ESP8266] : WiFi / HTTP
@enduml
```

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6. Activity Diagram (PlantUML)

```
@startuml
start
:Initialize Peripherals (ADC, I2C, PWM, UART);
:Calibrate Servos to Home Position;
repeat
  :Read 4x LDR Values;
  :Apply Predictive Filter to Reduce Noise;
  :Calculate Error (Horizontal & Vertical);
  if (Error > Threshold?) then (Yes)
    :Adjust PWM for Servos (Dual-Axis);
  endif
  :Read Power Data from INA219;
  :Update ThingSpeak via ESP8266;
  :Enter Low-Power Sleep (Short Duration);
repeat while (System Running?)
stop
@enduml
```

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7. Project Demo Setup

Hardware Requirements

- STM32F103 Development Board
- 2x SG90/MG90S Servo Motors (For vertical and horizontal movement)
- 4x LDR Sensors
- ESP8266 (NodeMCU veya 01S)
- INA219 High-Side DC Current Sensor

Software Requirements

- STM32CubeIDE (As development environment)
- FreeRTOS (For data management, CMSIS-RTOS integration)
- ThingSpeak Dashboard (For data visualization)

Demo Steps

1. System Startup: Power is applied and servo motors are moved to their starting position.
2. Light Tracking: LDRs are lighted by a torch. A system moving the motors in such a direction that it gets the light with a right angle, is observed.
3. Data Monitoring: Instant current and voltage data that are exported from ThingSpeak Dashboard are shown in graph.
4. Failure Scenario (Jitter-free Control): It is tested whether the system could still continue to work when one of sensors is closed.

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8. Risks & Mitigation

Risk	Probability	Impact	Mitigation Strategy
Servo Jitter	High	Medium	Implement Dead-band logic in PWM control.
Power Spikes	Medium	High	Use external power for Servos; common ground with STM32.
WiFi Connection Loss	Medium	Low	Implement retry logic for ESP8266 AT commands.
LDR Non-linearity	High	Medium	Use multi-point calibration and predictive filtering.

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9. Conclusion

This report outlines the planned development of **[Intelligent Dual-Axis Solar Tracker with Predictive Filtering]**, including:
✔ Task distribution among team members.
✔ Key milestones with deadlines.
✔ Quality assurance strategies.
✔ UML diagrams for system design.

Next Steps: Begin prototyping the sensor interface module.

Appendices

- [A] Datasheets (STM32F103, Sensors)
- [B] GitHub Repository Link: https://github.com/ozgeozler93/STM32F410_PhD_Projects/tree/main/02_Solar_Tracker_Project
- [C] Meeting Minutes (Weekly Updates)
